

Ex.no:2	EXPLORATORY DATA ANALYSIS (EDA) – DATA IMPORT AND EXPORT

Aim:

To inspect dataset features, detect null values, and export data into CSV, Excel, and SQLite database formats.

Algorithm:

1. Import pandas and sqlite3 libraries.
2. Read Life Expectancy Data.csv using `pd.read_csv()`.
3. Inspect data structure using `.head()`, `.tail()`, `.info()`, `.describe()`, and `.isnull().sum()`.
4. Export DataFrame to .csv and .xlsx formats using `.to_csv()` and `.to_excel()`.
5. Connect to Life Expectancy.db using sqlite3 and write data into an SQL table via `.to_sql()`.
6. Close the database connection and end execution.

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files
uploaded = files.upload()
file_name = next(iter(uploaded))
df = pd.read_csv(file_name)
df.columns = df.columns.str.strip()
df.head()

# -----
# 1. TOP 10 COUNTRIES BY AVERAGE LIFE EXPECTANCY
# -----

top_countries_life = (
df.groupby("Country")["Life expectancy"]
.mean()
.sort_values(ascending=False)
.head(10)
.reset_index()
)

print("Top 10 Countries by Average Life Expectancy")
display(top_countries_life)
plt.figure(figsize=(10,5))
plt.bar(top_countries_life["Country"], top_countries_life["Life
expectancy"])
```

```

plt.title("Top 10 Countries by Average Life Expectancy")
plt.xlabel("Country")
plt.ylabel("Average Life Expectancy")
plt.xticks(rotation=45)
plt.show()

# -----
# 2. TOP 10 COUNTRIES BY AVERAGE GDP
# -----

top_countries_gdp = (
df.groupby("Country") ["GDP"]
.mean()
.sort_values(ascending=False)
.head(10)
.reset_index()
)

print("Top 10 Countries by Average GDP")
display(top_countries_gdp)
plt.figure(figsize=(10,5))
plt.bar(top_countries_gdp["Country"], top_countries_gdp["GDP"])
plt.title("Top 10 Countries by Average GDP")
plt.xlabel("Country")
plt.ylabel("Average GDP")
plt.xticks(rotation=45)
plt.show()

# -----
# 3. TOP 10 COUNTRIES BY AVERAGE SCHOOLING
# -----

top_countries_schooling = (
df.groupby("Country") ["Schooling"]
.mean()
.sort_values(ascending=False)
.head(10)
.reset_index()
)

print("Top 10 Countries by Average Schooling")
display(top_countries_schooling)
plt.figure(figsize=(12,5))

```

```

plt.bar(top_countries_schooling["Country"],
top_countries_schooling["Schooling"])

plt.title("Top 10 Countries by Average Schooling")

plt.xlabel("Country")

plt.ylabel("Average Schooling")

plt.xticks(rotation=45)

plt.show()

# -----

# 4. DEVELOPMENT STATUS DISTRIBUTION

# -----

status_type = (
df["Status"]
.value_counts()
.reset_index()
)

status_type.columns = ["Status", "Count"]
print("Development Status Distribution")
display(status_type)

plt.figure(figsize=(8,5))
plt.bar(status_type["Status"], status_type["Count"])

plt.title("Development Status Distribution")

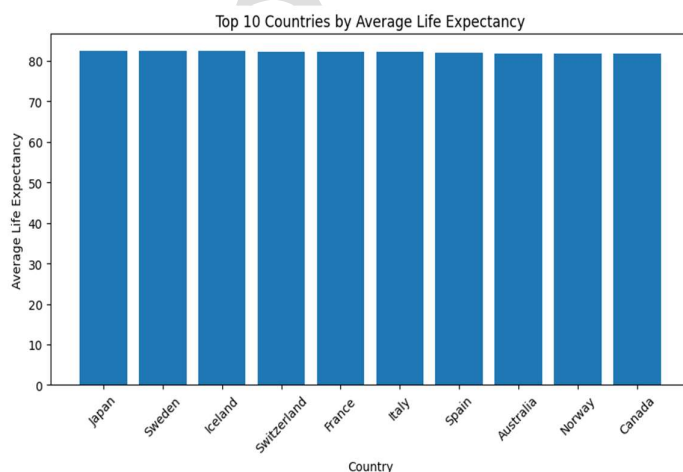
plt.xlabel("Status")

plt.ylabel("Number of Records")

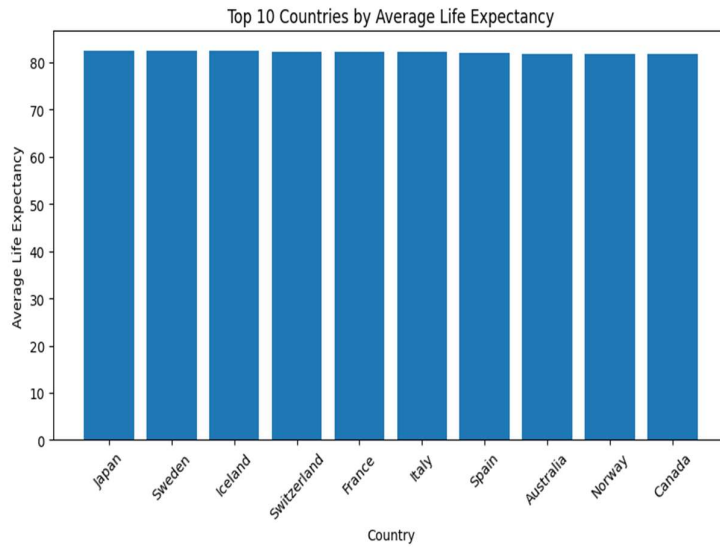
plt.show()

```

Output:

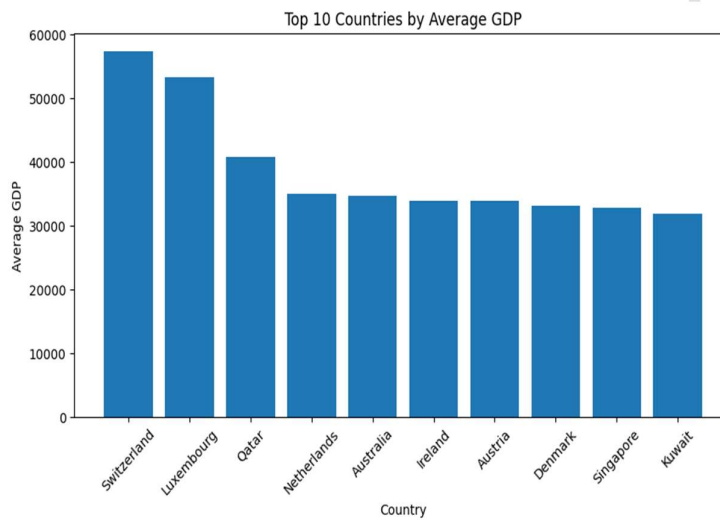


	Country	Life expectancy
0	Japan	82.53750
1	Sweden	82.51875
2	Iceland	82.44375
3	Switzerland	82.33125
4	France	82.21875
5	Italy	82.18750
6	Spain	82.06875
7	Australia	81.81250
8	Norway	81.79375
9	Canada	81.68750



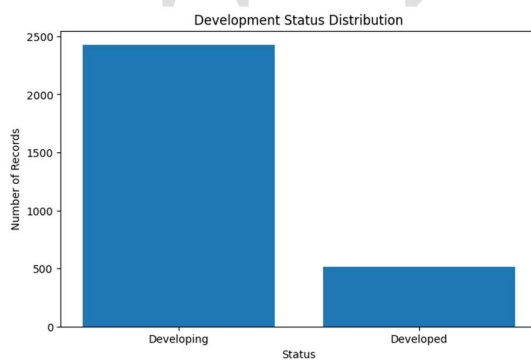
Top 10 Countries by Average GDP

	Country	GDP
0	Switzerland	57362.874601
1	Luxembourg	53257.012741
2	Qatar	40748.444104
3	Netherlands	34964.719797
4	Australia	34637.565047
5	Ireland	33835.272005
6	Austria	33827.476309
7	Denmark	33067.407916
8	Singapore	32790.105907
9	Kuwait	31914.378339



Top 10 Countries by Average Schooling

	Country	Schooling
0	Australia	20.03750
1	New Zealand	18.86875
2	Iceland	18.15625
3	Ireland	17.65625
4	Norway	17.46875
5	Finland	17.29375
6	Denmark	17.19375
7	Netherlands	17.05625
8	Belgium	16.78750
9	Germany	16.60000



Development Status Distribution

	Status	Count
0	Developing	2426
1	Developed	512

Result: